

Finding Borders

Time limit: 1.00 s

Memory limit: 512 MB

A *border* of a string is a prefix that is also a suffix of the string but not the whole string. For example, the borders of `abcababcbab` are `ab` and `abcbab`.

Your task is to find all border lengths of a given string.

Input

The only input line has a string of length n consisting of characters `a–z`.

Output

Print all border lengths of the string in increasing order.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
abcbababcbab	2 5